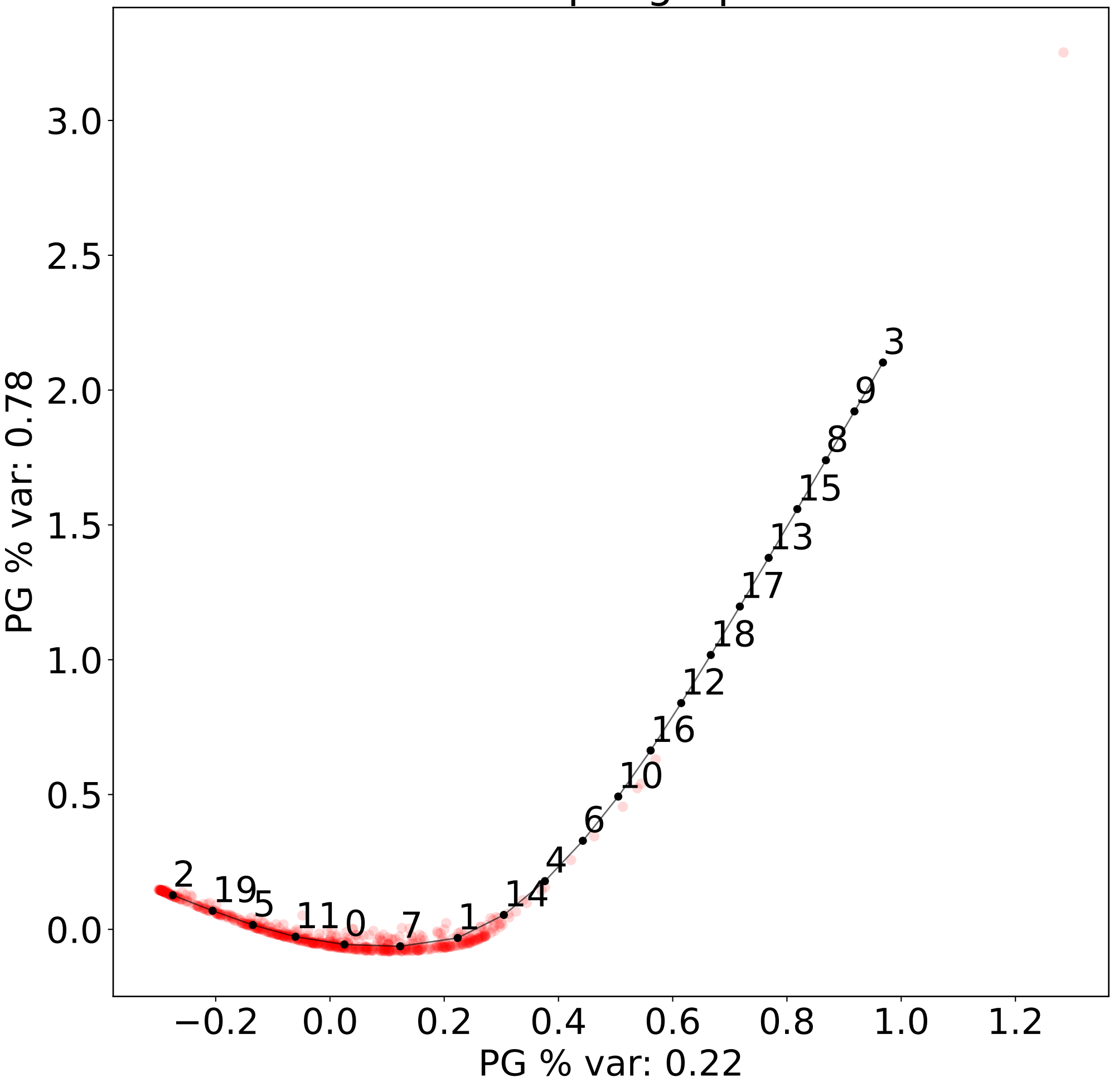Principal graph